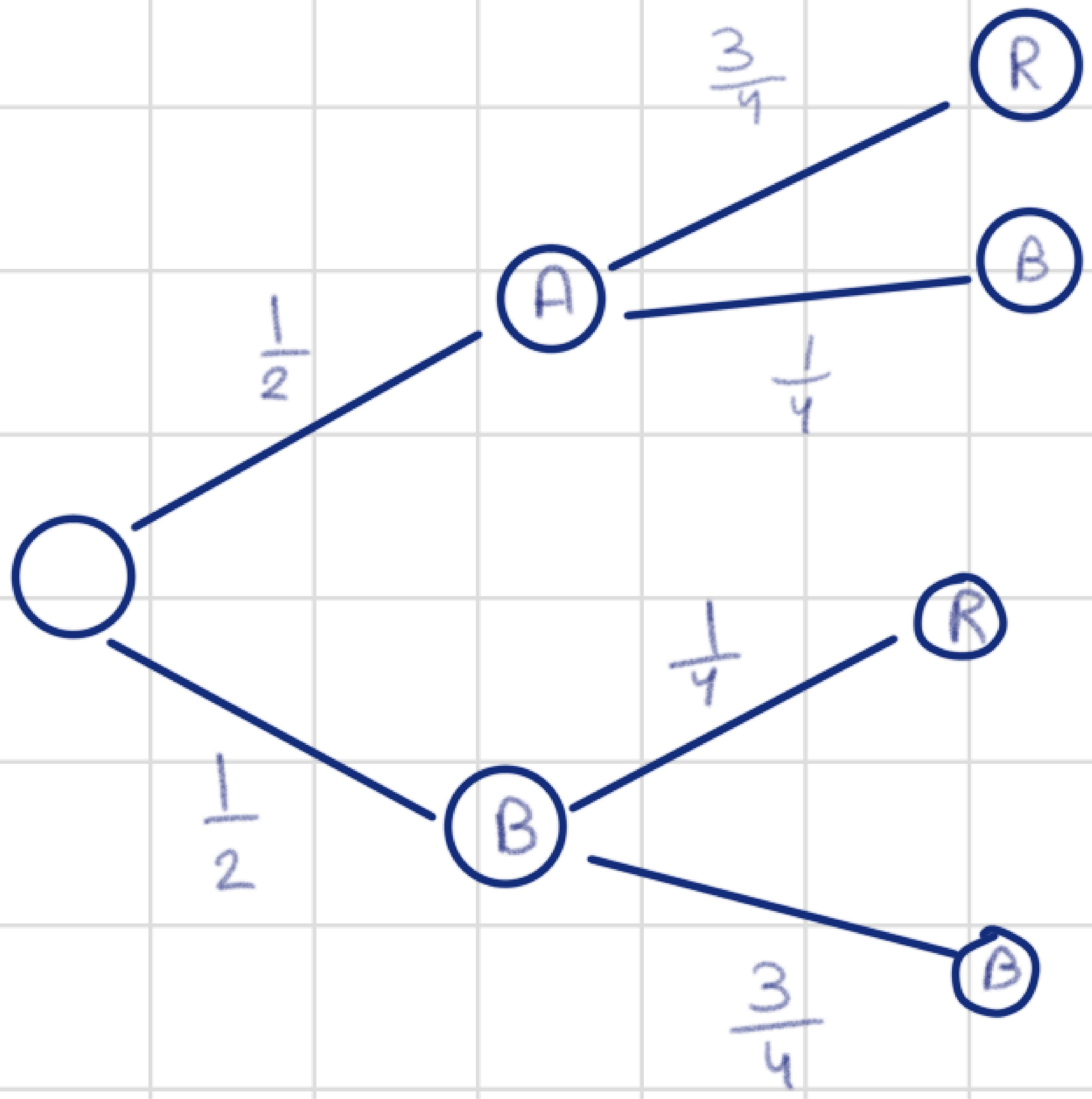
Bayes Theorem



$\longrightarrow$ Red ball

Bag A $\longrightarrow \frac{1}{2} * \frac{3}{4} = \frac{3}{8}$

Bag B $\longrightarrow \frac{1}{2} * \frac{1}{4} = \frac{1}{8}$

$$P(R) = \frac{3}{8} + \frac{1}{8} = \frac{4}{8}$$

$$P(A \cap R) = \frac{3}{8}$$

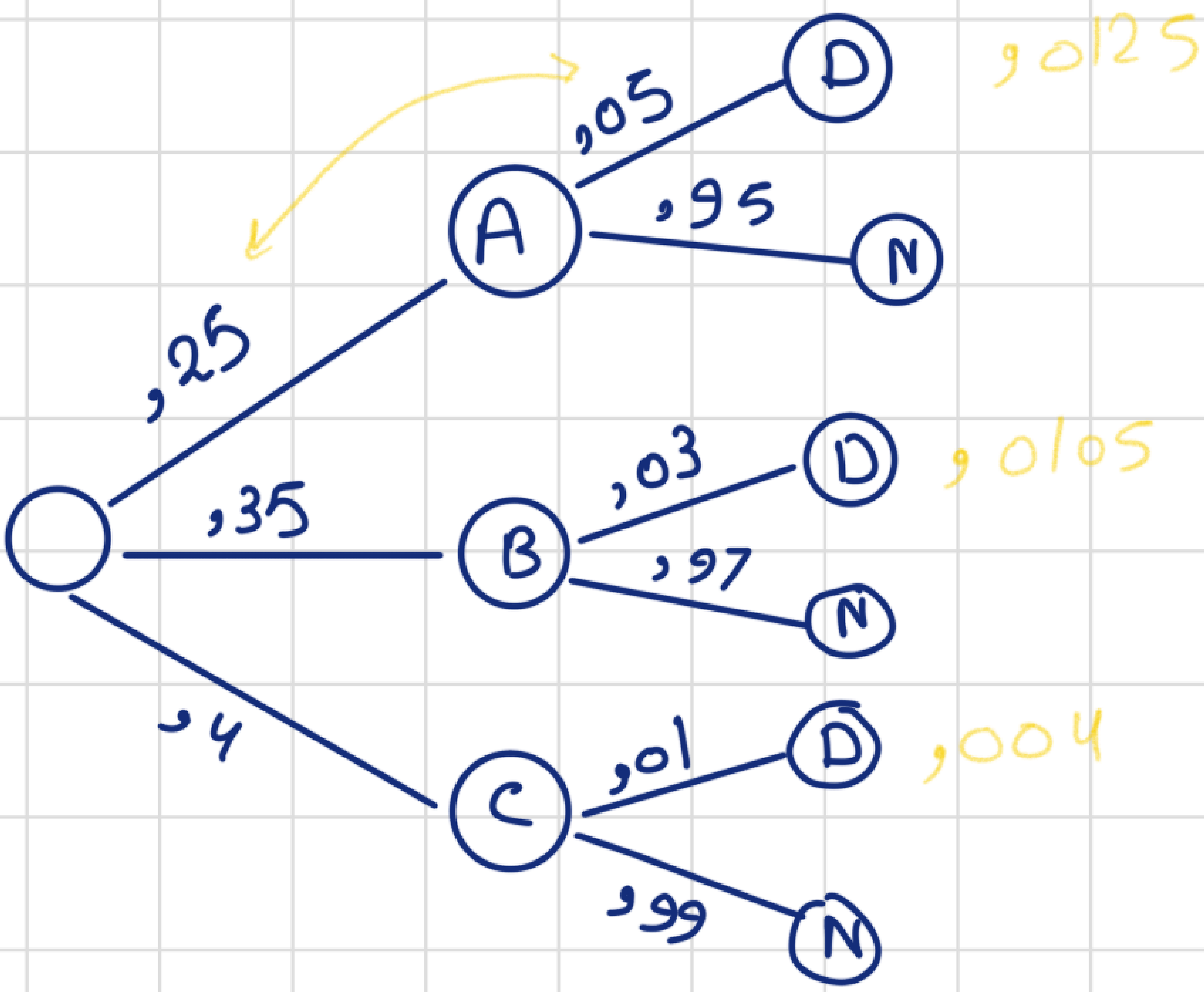
$$P(A|R) = \frac{\frac{3}{8}}{\frac{4}{8}} = \frac{3}{4}$$

$$P(A|B) = \frac{P(B|A)\,P(A)}{P(B)}$$

$$P(A|R) = \frac{P(R|A)\,P(A)}{P(R)}$$

→ Three events

A	B	C	
25%	35%	40%	
↓	↓	↓	
5%	3%	1%	→ defective



$$P(D) = ,027$$

$$P(C/D) = \frac{,004}{,027} = 14,8\%$$

Random variables

The probability of getting a certain outcome

associated with random processes

an event or experiment that has a random outcome

rolling a die, choosing a card.

it is something you can't exactly predict an outcome for

→ range of possibilities.

Random variables give numbers to outcomes of randoms events.

Discrete random variables

→ countable number of possible values.
→ probability of each value between 0 and 1.

Continous random variables

→ infinite number of possible values
→ probability of each distinct value is zero.

$M_x = \Sigma x_i p_i$ → mean of discrete random variable

$\sigma^2 = \Sigma (x_i - M)^2 F(x)$

variance of discrete random variable

Binomial Random Variable

count of the number of successes in a binomial experiment

Conditions

- fixed sample size (a certain number of trials)
- for each trial, The success must either happen or it must not
- probability of each event must be the same
- each trial must be an independent event .